
Simulations for Harnessing the VO₂ Phase Transition for Automatic Gain Control in Transimpedance Amplifiers

Gabriel Berezovsky

Supervised by

Amir Gildor and Prof. Yoav Kalcheim

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1 Yuanhang-based current-drive validation

This run checks that the Yuanhang-based implementation can produce a stable current-driven oscillation with a nonzero voltage valley. It validates the simulation mechanism; it is not a fit to the experimental sample.

All values were taken from Zhang *et al.* [1], except C_{th} . At 600 μA , the nominal value (49.6278 pJ/K) did not sustain oscillations, so it was increased fourfold to slow the thermal response and obtain a stable validation cycle. This control value is not an estimate of our sample's thermal capacitance.

Quantity	Value
Current pulse	600 μA from 2 to 38 μs
Simulation window	0–40 μs
Time step	0.5 ns
Initial state	$V(0) = 0 \text{ V}$, $T(0) = 324.9 \text{ K}$, insulating branch
Electrical capacitance	$C = 145.34619293 \text{ pF}$
Thermal capacitance	$C_{\text{th}} = 198.51107324 \text{ pJ/K}$ ($4\times$ control)
Environmental conductance	$S_e = 0.20558726 \text{ mW/K}$
Ambient temperature	$T_0 = 325 \text{ K}$
Metallic resistance	$R_m = 1286.3165 \Omega$
Noise and thermal neighbor coupling	zero
Analysis interval	10–38 μs

Table 1: Parameters used for the validation run. All unlisted resistance and hysteresis parameters retain their Yuanhang values.

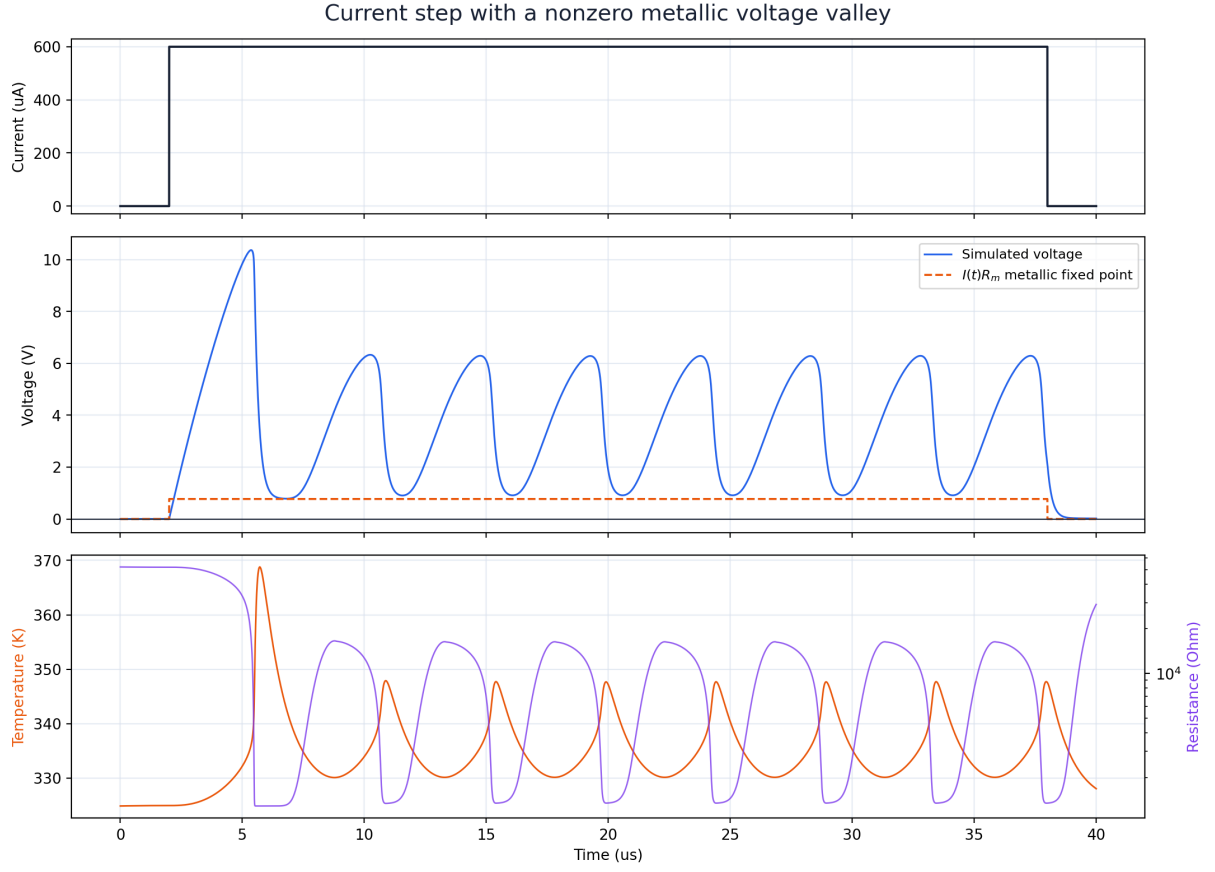


Figure 1: Applied current and simulated voltage, temperature, and resistance. The dashed orange line is the instantaneous metallic fixed point $I(t)R_m$. The analysis excludes the initial charging transient and uses the stable interval from 10 to 38 μs .

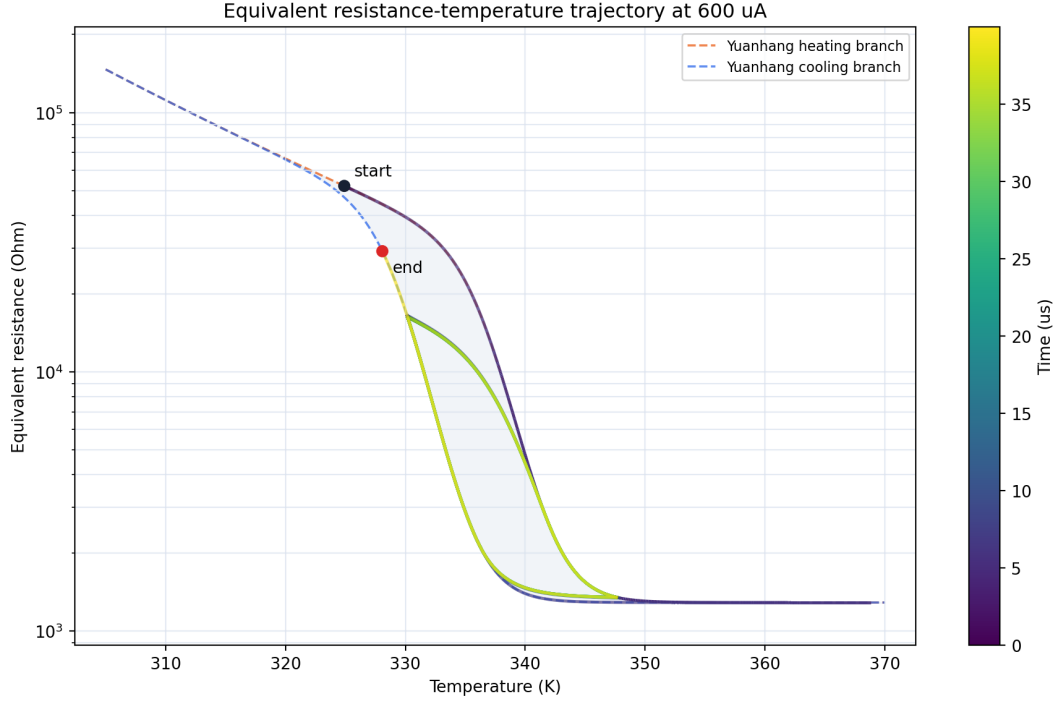


Figure 2: Equivalent resistance versus temperature. The dashed curves and shaded band show Yuanhang’s major heating and cooling hysteresis; color shows progression through the 40 μs simulated trajectory.

Metric	Result
Oscillation detected	yes
Detected cycles	5
Frequency	0.221685 MHz
Period	4.51090 μs
Period coefficient of variation	1.77×10^{-4}
Steady voltage range	0.905654–6.33281 V
Full-pulse temperature range	324.9–368.8 K
Steady temperature range	330.16–347.91 K
Steady resistance range	1.339–16.382 k Ω
Metallic electrical time CR_m	186.961 ns
Thermal time C_{th}/S_e	0.965581 μs

Table 2: Metrics measured from the validation run.

The metallic-state voltage is set approximately by $V = IR_m$:

$$V_{\text{metal}} \approx IR_m = (600 \times 10^{-6} \text{ A})(1286.3165 \Omega) = 0.771790 \text{ V}. \quad (1)$$

The simulated minimum is 0.905654 V; it remains above this fixed point because switching reverses before the capacitor settles. The experimental manuscript reports 40–60 MHz oscillations [2], whereas this validation gives 0.221685 MHz. Sample-specific resistance and time constants are therefore still required for a quantitative match.

2 Sample-specific resistance–temperature fit

The supplied file contains 264 measurements from one cooling sweep and one heating sweep over 289.85–370.16 K at 2 μ A. We fitted the major-loop form

$$R(T, \delta) = R_m + R_0 e^{(E_a/k_B)/T} \left[\frac{1}{2} + \frac{1}{2} \tanh\left(\beta \left(\delta \frac{w}{2} + T_c - T\right)\right) \right], \quad \delta = \begin{cases} -1, & \text{cooling,} \\ +1, & \text{heating.} \end{cases} \quad (2)$$

The fit minimizes the residual in $\log_{10} R$. The minor-loop parameter γ was fixed to Yuanhang’s value because a major loop alone does not determine it.

Here R_0 is the Arrhenius prefactor, not the insulating resistance. On the fully insulating branch,

$$R_{\text{ins}}(T) \approx R_m + R_0 e^{(E_a/k_B)/T}. \quad (3)$$

For example, the fitted sample gives $R_{\text{ins}}(300 \text{ K}) \approx 3.78 \text{ k}\Omega$, well above the metallic floor $R_m = 18.22 \Omega$.

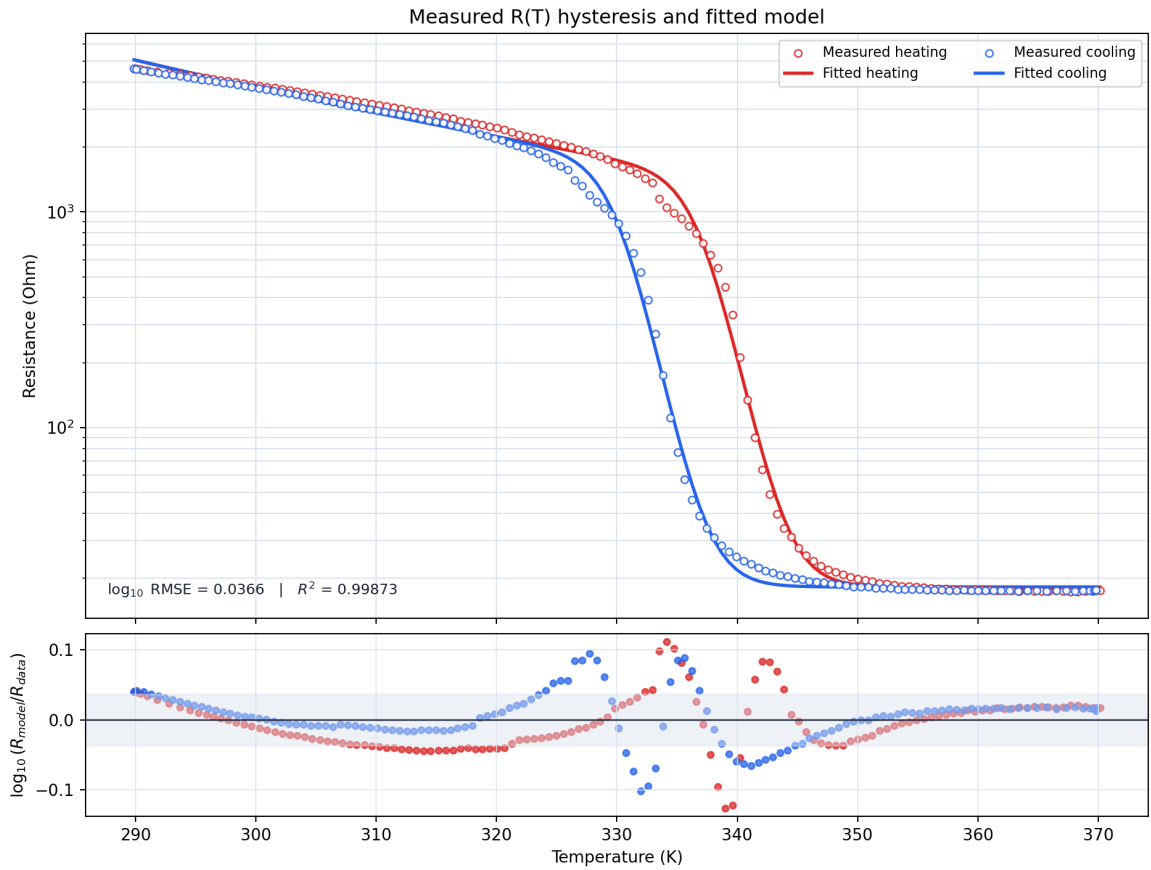


Figure 3: Measured heating and cooling branches, fitted model, and log-resistance residuals.

Parameter	Estimate	95% interval	Yuanhang
R_0 (prefactor)	0.7989 Ω	0.3774–1.8483 Ω	0.005359 Ω
E_a/k_B	2536.97 K	2272.55–2767.52 K	5220.47 K
E_a	0.21862 eV	0.19583–0.23849 eV	0.44987 eV
R_m (metallic floor)	18.2153 Ω	17.4943–18.9010 Ω	1286.32 Ω
T_c	333.494 K	333.137–333.838 K	332.806 K
w	6.8823 K	6.6291–7.2681 K	7.1936 K
β	0.29916 K ⁻¹	0.28092–0.32021 K ⁻¹	0.25280 K ⁻¹
$T_{\text{cool}} = T_c - w/2$	330.052 K	329.642–330.385 K	329.209 K
$T_{\text{heat}} = T_c + w/2$	336.935 K	336.534–337.348 K	336.403 K
γ	0.95627 (fixed)	—	0.95627

Table 3: Sample-specific resistance parameters compared with Yuanhang’s reference implementation, which uses the effective metallic resistance $R_m = 1286.32 \Omega$. R_0 and E_a/k_B are strongly correlated; their joint combination determines the insulating branch.

The fit gives $R^2 = 0.998734$ and a $\log_{10} R$ RMSE of 0.03663, equivalent to a typical multiplicative error of 1.088. The largest deviation is a factor of 1.336 inside the switching region; this residual is retained rather than adding parameters unsupported by the measurement.

3 Experimental timing and ambient temperature

The sample-specific simulations will reproduce the measured pulse timing rather than use an instantaneous current step. The TIA sweep is taken to share the ambient range reported for the related measurements [2].

Input	Value
Ambient temperature	$T_0 \approx 314.4$ K (314.25–314.55 K)
Pulse width	300 ns
Current rise time	26–27 ns (10–90%)
Current fall time	18–23 ns (90–10%)

Table 4: Measured environmental and timing inputs retained for later simulation of the experiment.

4 Environmental thermal conductance

The objective is to estimate the environmental thermal conductance S_e , which sets the heat-loss rate $S_e(T - T_0)$. The estimate requires a hot operating point whose temperature is approximately stationary. We use the following procedure:

1. Order the measured records by their baseline-corrected input current and identify the last record without coherent oscillation. This gives the stable point closest to switching, where the temperature rise is large but a quasi-steady balance remains defensible.
2. Subtract the pre-pulse current and voltage offsets and select a settled window in which V/I is nearly constant. In this window $dV/dt \approx 0$, so the measured input current is approximately the resistive current through the VO₂ device.

3. Calculate the effective resistance $R_* = V_*/I_*$ and Joule power $P_* = I_*V_*$. Invert the fitted heating branch, $R_{\text{heat}}(T_*) = R_*$, to infer the device temperature T_* .
4. Apply the quasi-steady thermal balance $C_{\text{th}} dT/dt = P_* - S_e(T_* - T_0) \approx 0$, which gives $S_e = P_*/(T_* - T_0)$.

The filenames identify the nominal source-voltage setting, not the voltage across the VO₂ device and not the simulation input. We therefore retain that setting only for traceability and use the measured current. Over the common 50–250 ns analysis window, the 250 and 300 mV source settings produced baseline-corrected current steps of 189.6 and 228.2 μA , respectively.

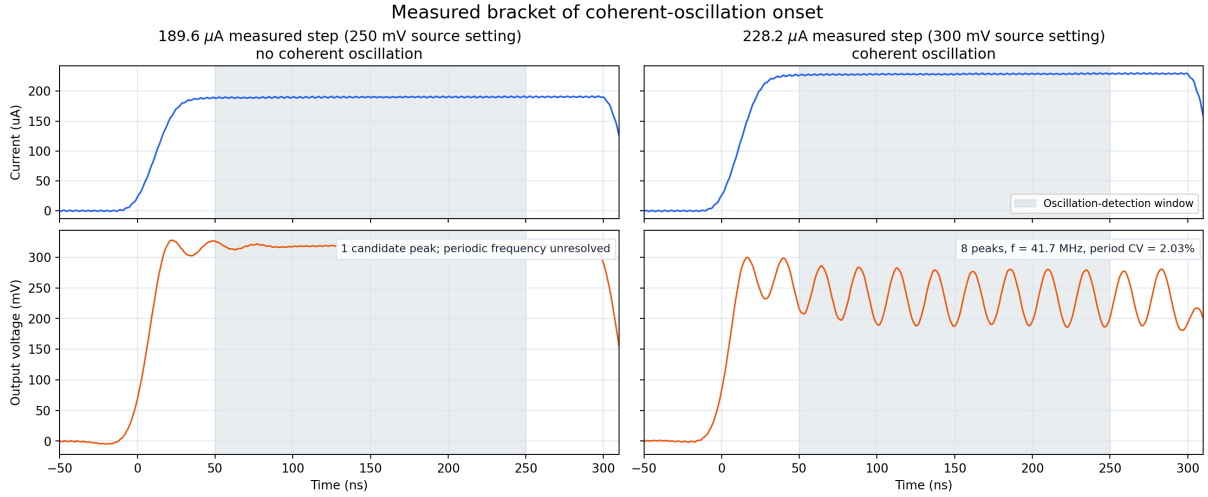


Figure 4: Adjacent baseline-corrected records bracketing oscillation onset on shared axes. The 189.6 μA measured current step (250 mV source setting) contains only one candidate peak, so no periodic frequency is assigned. At 228.2 μA (300 mV source setting), eight peaks give 41.7 MHz with a 2.03% period coefficient of variation.

Figure 4 implements the first step of the procedure. The 228.2 μA record is the first trace with coherent oscillation, while the immediately preceding 189.6 μA record does not oscillate. We therefore selected the latter as the closest measured point below switching onset. We did not use the first transition during the current edge because the device is still heating there and $dT/dt = 0$ would not be justified.

For the selected trace, the current and voltage offsets are their medians from -200 to -50 ns. We then use the settled 100–250 ns interval. Its effective resistance changes by only 0.145%, supporting the quasi-steady approximation. Figure 5 shows the waveform, the stability test, and the temperature inversion.

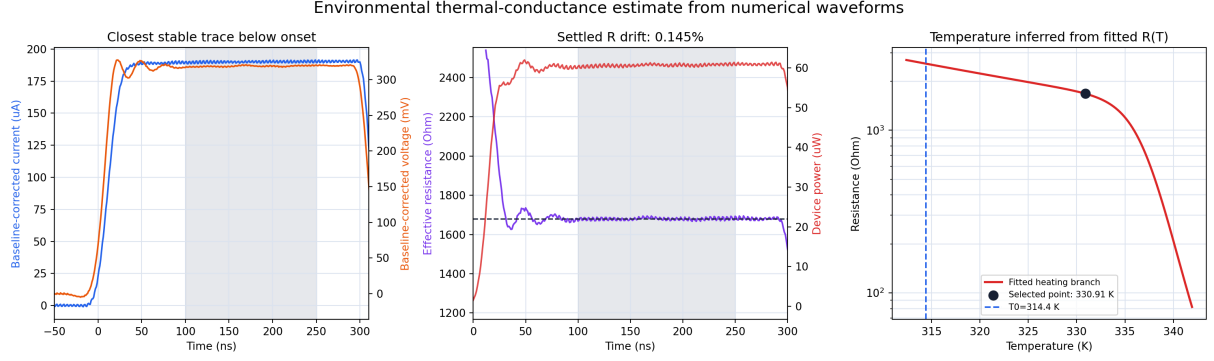


Figure 5: Environmental-conductance evidence. Gray shading marks the settled interval. The measured $R_* = V_*/I_*$ is mapped to T_* through the fitted heating branch.

The baseline-corrected medians give

$$I_* = 190.162 \mu\text{A}, \quad V_* = 319.213 \text{ mV}, \quad R_* = \frac{V_*}{I_*} = 1679.759 \Omega, \quad (4)$$

and

$$P_* = I_* V_* = 60.658 \mu\text{W}. \quad (5)$$

Solving the fitted heating branch $R_{\text{heat}}(T_*) = R_*$ gives $T_* = 330.905 \text{ K}$. At a quasi-steady point, the thermal equation reduces to

$$C_{\text{th}} \frac{dT}{dt} = P_* - S_e(T_* - T_0) \approx 0, \quad S_e = \frac{P_*}{T_* - T_0}. \quad (6)$$

Therefore,

$$S_e = \frac{60.658 \mu\text{W}}{330.905 \text{ K} - 314.4 \text{ K}} = 3.675 \mu\text{W/K} = 0.003675 \text{ mW/K}. \quad (7)$$

Result	Value
Selected trace	190.162 μA settled current (250 mV setting)
First oscillating trace	228.2 μA current step (300 mV setting)
Settled resistance drift	0.145%
Inferred temperature	330.905 K
Temperature rise	16.505 K
S_e estimate	0.003675 mW/K
Conditional 95% interval	0.003434–0.004085 mW/K
Yuanhang reference	0.205587 mW/K

Table 5: Environmental thermal-conductance calculation and comparison.

The interval propagates paired waveform block resampling, the 1000-sample $R(T)$ fit bootstrap, and the 314.25–314.55 K ambient range. Omitting baseline correction gives 0.003560 mW/K, inside the interval. The result remains conditional on the $R(T)$ curve and TIA traces belonging to the same device and on the quasi-static heating branch applying under electrical drive. Yuanhang’s conductance is about 56 times larger.

5 Electrical capacitance

The objective is to estimate the effective electrical capacitance C , which stores charge at the device node and sets the electrical response time $\tau_{\text{el}} = R(T)C$. In the ideal-current model, the measured input current splits between the VO₂ resistance and this capacitance:

$$I_{\text{in}} = \frac{V}{R(T)} + C \frac{dV}{dt}. \quad (8)$$

We therefore seek the part of the measured current that cannot be explained by the instantaneous resistive current $V/R(T)$. The analysis uses low-current, non-switching rising edges so that the device begins on the known cold heating branch and a phase transition does not complicate the electrical transient. It assumes that the current and voltage channels are time-aligned, their pre-pulse offsets are removable, C is approximately constant over the fit window, and the reduced circuit contains the relevant current paths. The procedure is:

1. Subtract each trace’s pre-pulse current and voltage medians and select the initial rising edge of the 100–250 mV source-setting records, all of which remain below switching.
2. Use $C_{\text{shortcut}} = \Delta I / (dV/dt)$ as a preliminary scale estimate. This expression is valid only if the resistive current is negligible; it must not be accepted before checking the full current balance.
3. Evaluate the physically complete residual $I_C = I_{\text{in}} - V/R(T)$ using the fitted cold resistance, and fit $I_C = C dV/dt$ over the early edge with the physical constraint $C \geq 0$.
4. Check the result independently against the measured current–voltage timing. If no positive lag is resolved, convert the timing resolution Δt into an upper bound $C \lesssim \Delta t / R(T_0)$. We adopt that bound as a conservative modeling value while retaining $C = 0$ as the best-fit lower endpoint.

We first reproduce the preliminary shortcut. For each trace, ΔI is the baseline-to-plateau current step and dV/dt is a linear fit over 0–30 ns, giving

$$C_{\text{shortcut}} = \frac{\Delta I}{dV/dt}, \quad \frac{\mu\text{A}}{\text{mV/ns}} = \text{pF}. \quad (9)$$

Drive setting	ΔI (μA)	dV/dt (mV/ns)	C_{shortcut} (pF)
100 mV	76.171	3.562	21.382
150 mV	114.172	6.127	18.634
200 mV	152.162	8.235	18.477
250 mV	189.601	9.008	21.047
Median			19.841

Table 6: Reconstruction of the original 19.8 pF shortcut estimate.

The median shortcut estimate is 19.8 pF, but it is biased because the measured current itself rises over 26–27 ns and most of it flows resistively. Applying the full current balance gives

$$C \frac{dV}{dt} = I_{\text{in}} - \frac{V}{R(T)}, \quad C = \frac{I_{\text{in}} - V/R(T)}{dV/dt}. \quad (10)$$

At $T_0 = 314.4$ K, the fitted heating branch gives $R(T_0) = 2.570$ k Ω . Substituting the measured current and voltage leaves no resolved capacitive current: the unconstrained early-edge fit is approximately -0.25 to -0.10 pF, depending on the fit window, and the physical nonnegative optimum is therefore $C = 0$. The current and voltage 50% crossings differ by only -0.74 to $+0.52$ ns. Taking 1 ns as the timing resolution gives the non-statistical bound

$$C \lesssim \frac{1 \text{ ns}}{2.570 \text{ k}\Omega} \approx 0.39 \text{ pF}. \quad (11)$$

In contrast, 19.8 pF would give $R(T_0)C = 50.9$ ns and a large voltage lag that is absent from the data. A larger capacitance would worsen the disagreement. This does not establish that the physical capacitance is exactly zero; it means that a positive value is not identifiable from the present traces. To avoid assuming an instantaneous electrical response, the sample-specific simulations and downstream thermal fit use the conservative upper-bound value

$$C_{\text{adopt}} = 0.39 \text{ pF}, \quad I_R = I_{\text{in}} - C_{\text{adopt}} \frac{dV}{dt}. \quad (12)$$

This is not a measured central estimate: $C = 0$ remains the constrained best fit, and 0.39 pF intentionally maximizes the electrical memory still compatible with the approximately 1 ns timing resolution.

This conclusion is independent of the present choice of γ . The capacitance fit uses the initial cold heating edge, before a temperature reversal; γ modifies only minor hysteresis loops after a reversal and therefore does not enter this calculation. Changing γ can alter later oscillation shapes and frequencies, but it cannot supply the missing early-edge voltage lag. A numerical nonzero estimate requires a faster, low-amplitude nonswitching edge with calibrated current-voltage channel timing; with R approximately constant, $C = \tau_{\text{el}}/R$.

6 Thermal time constant and thermal capacitance

The objective is to estimate the effective thermal capacitance C_{th} , which relates net heating power to the device's temperature rate of change. Together with the environmental conductance found in Section 4, it sets the thermal response time

$$\tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \quad (13)$$

A larger C_{th} means that more energy is required to change the device temperature and therefore produces a slower thermal response. Section 5 found no resolved positive electrical capacitance but established the upper bound $C_{\text{adopt}} = 0.39$ pF. We use this conservative value and first separate the resistive current from the capacitive current. The fitted heating branch then acts as a thermometer:

$$I_R(t) = I_*(t) - C_{\text{adopt}} \frac{dV_*}{dt}, \quad R_*(t) = \frac{V_*(t)}{I_R(t)}, \quad T_*(t) = R_{\text{heat}}^{-1}(R_*(t)). \quad (14)$$

The estimate assumes a single lumped device temperature, constant C_{th} and S_e over the fitted interval, the same sample for the electrical and $R(T)$ measurements, and validity of the fitted heating branch before any thermal reversal. It is conditional on the adopted 0.39 pF upper bound and requires baseline-corrected, time-aligned current and voltage channels. The procedure is:

1. Select moderate non-switching traces with sufficient signal, subtract their pre-pulse offsets, and smooth the oscilloscope noise before evaluating dV_*/dt and forming V_*/I_R . We use measured current steps of 76.2, 114.2, and 152.2 μA (the 100, 150, and 200 mV source settings); the 189.6 μA record (250 mV setting) is reserved as a near-transition sensitivity check.
2. Compute $I_R = I_* - C_{\text{adopt}}dV_*/dt$, reconstruct the measured temperature trajectory from $R_*(t) = V_*(t)/I_R(t)$ and the inverse fitted heating branch, and calculate the applied Joule power $P_*(t) = I_R(t)V_*(t)$.
3. For trial values of C_{th} , integrate the thermal balance using the previously estimated S_e and T_0 . Fit one shared C_{th} by minimizing the temperature mismatch across all three primary traces over 15–35 ns, then obtain $\tau_{\text{th}} = C_{\text{th}}/S_e$.
4. Test robustness by fitting each trace separately, checking the excluded 189.6 μA record, varying the fit window, and propagating the $R(T)$, S_e , ambient temperature, and waveform uncertainties.

The thermal trajectory for each trial value is obtained from

$$C_{\text{th}} \frac{dT}{dt} = P_*(t) - S_e(T - T_0), \quad C_{\text{th}} = S_e \tau_{\text{th}}. \quad (15)$$

A nine-sample, second-order Savitzky–Golay filter is used in the reconstruction. Because the fit ends before a thermal reversal, γ does not enter this estimate. Figure 6 compares the temperatures inferred directly from the electrical measurements with the trajectories predicted by the shared thermal model.

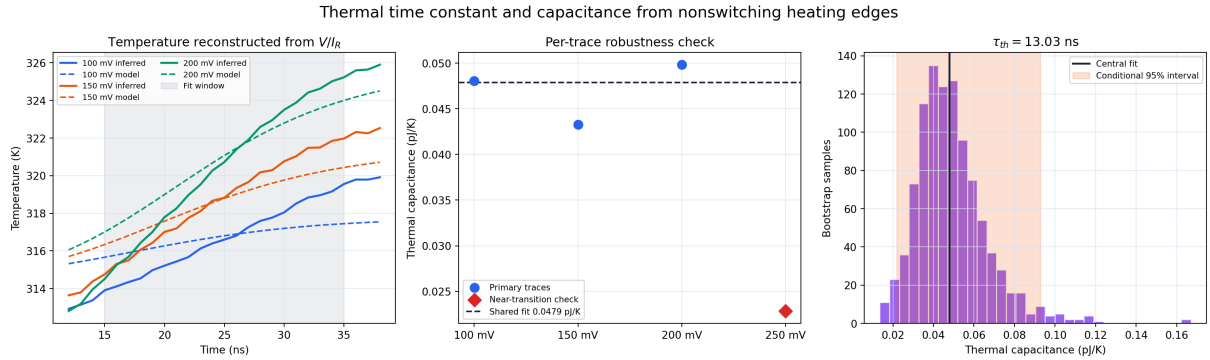


Figure 6: Thermal-capacitance evidence conditioned on $C_{\text{adopt}} = 0.39$ pF. Solid curves are temperatures inferred from V_*/I_R ; dashed curves are the shared thermal model. The 189.6 μA measured current step (250 mV source setting) is retained only as a near-transition sensitivity check.

Quantity	Result
Ambient temperature	314.4 K
Environmental conductance	0.00367513 mW/K
Adopted electrical capacitance	0.39 pF (upper bound)
Selected traces	76.2, 114.2, and 152.2 μ A steps
Fit interval	15–35 ns
Thermal time constant τ_{th}	13.026 ns
Conditional 95% interval for τ_{th}	5.865–25.174 ns
Thermal capacitance C_{th}	0.047873 pJ/K
Conditional 95% interval for C_{th}	0.021918–0.092624 pJ/K
Temperature-fit RMSE	1.145 K
Individual selected-trace range	0.043261–0.049830 pJ/K
189.6 μ A sensitivity result	0.022858 pJ/K; RMSE 2.380 K
Measured oscillation period	16–24 ns
Yuanhang reference	49.6278 pJ/K

Table 7: Thermal time-constant and capacitance results.

The conditional interval propagates trace resampling, the $R(T)$ bootstrap, the conductance bootstrap, the ambient-temperature range, and ± 2 ns changes to the fit window. The archived conductance samples do not retain their paired $R(T)$ parameter draws, so the two bootstraps are resampled independently; this is conservative but does not preserve their covariance. The result remains conditional on the adopted electrical-capacitance upper bound, the static heating branch describing the driven device, and the electrical and $R(T)$ measurements belonging to the same sample. Relative to the $C = 0$ analysis, the upper-bound correction raises the central C_{th} estimate by only 1.5% (0.047163 to 0.047873 pJ/K), although the RMSE increases from 0.806 to 1.145 K. The 189.6 μ A trace gives a smaller value and a substantially larger error because it approaches a reversal; including it would bias the single-heating-branch fit downward. The fitted 13.0 ns thermal time is of the same order as the measured 16–24 ns oscillation period. Yuanhang’s thermal capacitance is approximately 1037 times larger.

The experimental manuscript reports a 150 nm film thickness and an approximately 200 nm electrode gap [2], but not the electrically active filament width or volume. A numerical geometry estimate $C_{\text{th}} = \rho c_p V$ would therefore introduce an unconstrained volume and is not reported.

7 Comparison across experimental currents

This is a blind prediction test: the fitted $R(T)$ curve, $T_0 = 314.4$ K, $S_e = 0.00367513$ mW/K, $C = 0.39$ pF, and $C_{th} = 0.047873$ pJ/K are fixed before examining the result. Each baseline-corrected measured current waveform is replayed through the model; no parameter is adjusted by current. Predicted voltages are sampled at the experimental 1 ns times, and both data sets use the same 50–250 ns window and oscillation detector.

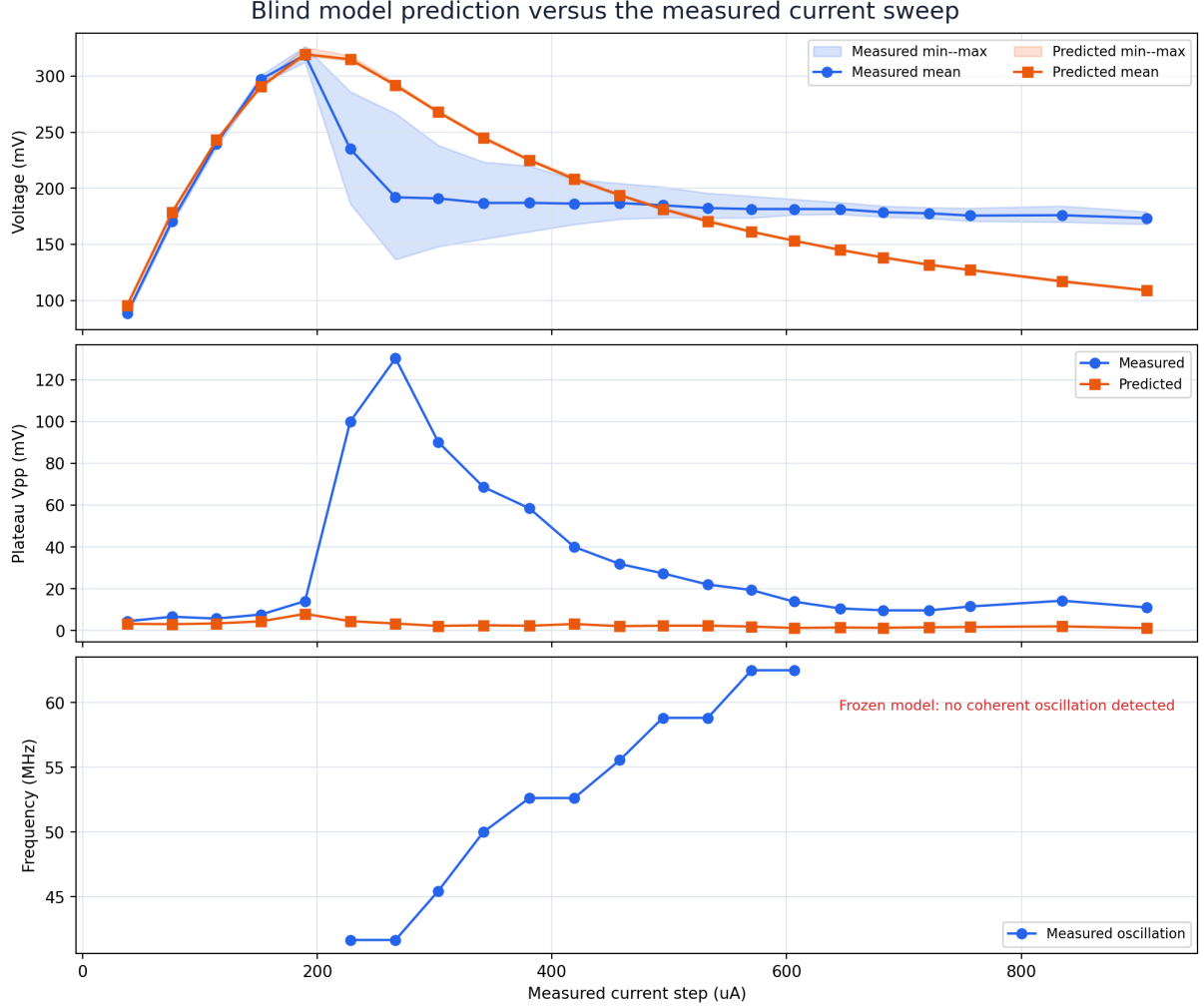


Figure 7: Measured behavior and blind model predictions over all 22 measured current steps. The shaded bands show the voltage extrema in the common analysis window.

The experiment oscillates over 11 records, from 228.2 to 606.3 μ A, with frequencies of 41.7–62.5 MHz. The frozen model predicts no coherent oscillation at any measured current. It nevertheless reproduces the last stable pre-onset record: at 189.6 μ A, the measured and predicted mean voltages are 318.86 and 319.53 mV. Thus the cold-side static calibration works, whereas the dynamic switching window does not.

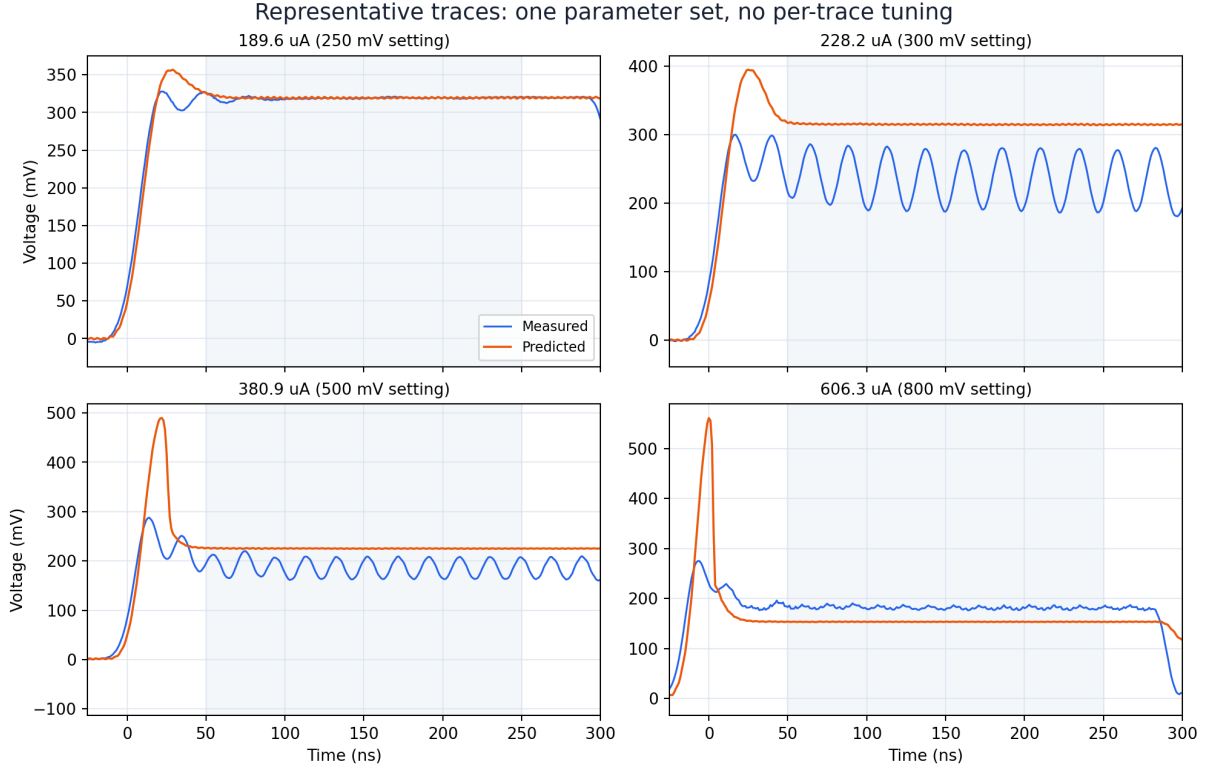


Figure 8: Representative measured and predicted voltages. The shaded interval is the common detection window. The 189.6 μA control is reproduced closely, but the measured oscillations above onset are replaced by steady model solutions.

Quantity	Experiment	Frozen model
Oscillating records	11 of 22	0 of 22
Oscillation-current interval	228.2–606.3 μA	none
Frequency interval	41.7–62.5 MHz	—
Energy per cycle in oscillating records	1.229–1.761 pJ	—
189.6 μA mean voltage	318.86 mV	319.53 mV
Median plateau-voltage RMSE	32.38 mV	
Maximum change after halving Δt	0.228 mV	

Table 8: Direct prediction results. Energy per cycle is the mean plateau power divided by the detected frequency.

Halving the integration step from 0.05 to 0.025 ns does not change any tested classification and changes the representative mean voltages by at most 0.228 mV. The missing oscillations are therefore not a time-step artifact.

8 Capacitance dependence and the thermal-only limit

In the zero-capacitance limit,

$$C = 0 \implies V(t) = I_{\text{in}}(t)R(T), \quad \tau_{\text{th}} = \frac{C_{\text{th}}}{S_e}. \quad (16)$$

A thermal relaxation cycle requires the same current to heat through the heating transition and then cool through the cooling transition. Using the fitted branch resistances at the two transition centers gives

$$I > I_{\text{heat}} = \sqrt{\frac{S_e(T_{\text{heat}} - T_0)}{R_{\text{heat}}}} = 329.65 \mu\text{A}, \quad I < I_{\text{cool}} = \sqrt{\frac{S_e(T_{\text{cool}} - T_0)}{R_{\text{cool}}}} = 254.44 \mu\text{A}. \quad (17)$$

Because $I_{\text{heat}} > I_{\text{cool}}$, these conditions do not overlap: the fitted system has no thermal-only oscillation window. At the adopted $C = 0.39$ pF, the metallic electrical time is $CR_m = 7.10$ ps, compared with $\tau_{\text{th}} = 13.026$ ns. The ratio is 5.45×10^{-4} , so this capacitance supplies negligible electrical memory and does not alter the steady thermal thresholds.

The numerical study varies C from 0 to 50 pF and C_{th} across its conditional 0.021918–0.092624 pJ/K interval while holding every other parameter fixed. The experimentally admissible range $0 \leq C \leq 0.39$ pF produces no oscillation anywhere in that interval. Oscillations first appear at 3 pF only at the lowest tested C_{th} ; at the adopted C_{th} , they first appear at 7 pF. Both values exceed the electrical timing bound and would create a voltage lag absent from the measured edge.

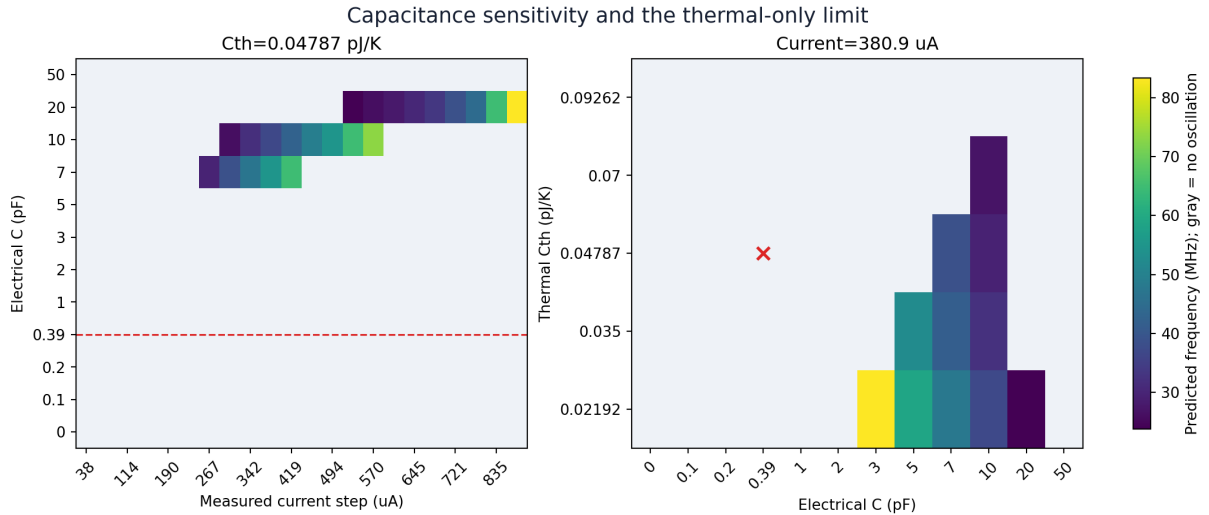


Figure 9: Predicted frequency under capacitance variation. Gray denotes no coherent oscillation. The dashed line and red cross identify the adopted electrical and thermal capacitances; larger electrical values are stress tests outside the timing bound.

The present parameter set therefore matches the stable pre-onset voltage but cannot reproduce the measured oscillatory regime without an electrical capacitance that contradicts the edge timing. The next useful tests are an independently measured dynamic switching loop and inclusion of the actual TIA/load impedance. Changing γ alone cannot explain the onset at $228.2 \mu\text{A}$, because the predicted temperature there does not reach the first heating transition; γ acts only after a hysteresis reversal.

9 Global parameter inference from all measured waveforms

We next asked whether one shared set of model parameters could be inferred directly from the complete current–voltage data set. This is a stricter test than fitting one trace: the same vector

$$\theta = (C, C_{\text{th}}, S_e, T_0, T_c, w, \beta, \gamma) \quad (18)$$

must describe all 22 measured current waveforms. The source settings at 200, 400, 600, 800, and 1000 mV were withheld from optimization, leaving 17 training traces and five blind test traces. The source-setting labels only identify records; the measured current waveform is the model input in every case.

For each candidate, the simulator uses the complete measured current transient and compares the predicted and measured voltage over the common 50–250 ns plateau. The weighted objective is

$$J = 0.20J_{\text{wave}} + 0.20J_{\text{mean}} + 0.15J_{\text{amp}} + 0.15J_{\text{spec}} + 0.15J_f + 0.10J_{\text{class}} + 0.05J_{\text{edge}} + 0.05J_{\text{prior}}. \quad (19)$$

Here J_{wave} is a normalized waveform error, with a ± 12 ns phase shift allowed only for oscillatory records; the other terms compare mean voltage, peak-to-peak amplitude, normalized spectrum, frequency, oscillation classification, and the 0–50 ns rising edge. The constrained fit includes a weak prior and keeps each parameter inside its independently estimated interval. A second, deliberately relaxed fit removes that prior and expands the bounds to test whether the model can fit the data only by abandoning those measurements. Differential evolution explores the global space, followed by bounded local refinement. Optimization uses $\Delta t = 0.2$ ns; every reported candidate is rerun at 0.05 ns.

Parameter	Starting value	Physical interval	Constrained fit	Relaxed fit
C (pF)	0.3900	0–0.390	0.2408	5.4187*
C_{th} (pJ/K)	0.04787	0.02192–0.09262	0.06146	0.10643*
S_e (mW/K)	0.003675	0.003434–0.004085	0.003621	0.004894*
T_0 (K)	314.400	314.25–314.55	314.416	317.844*
T_c (K)	333.494	333.137–333.838	333.476	332.927*
w (K)	6.8823	6.6291–7.2681	6.8726	2.5568*
β (K ^{−1})	0.29916	0.28092–0.32021	0.29542	0.11613*
γ	0.95627	0.2–2.0	0.97956	1.03546

Table 9: Shared parameters inferred from all training traces. An asterisk marks a relaxed value outside the independently supported physical interval.

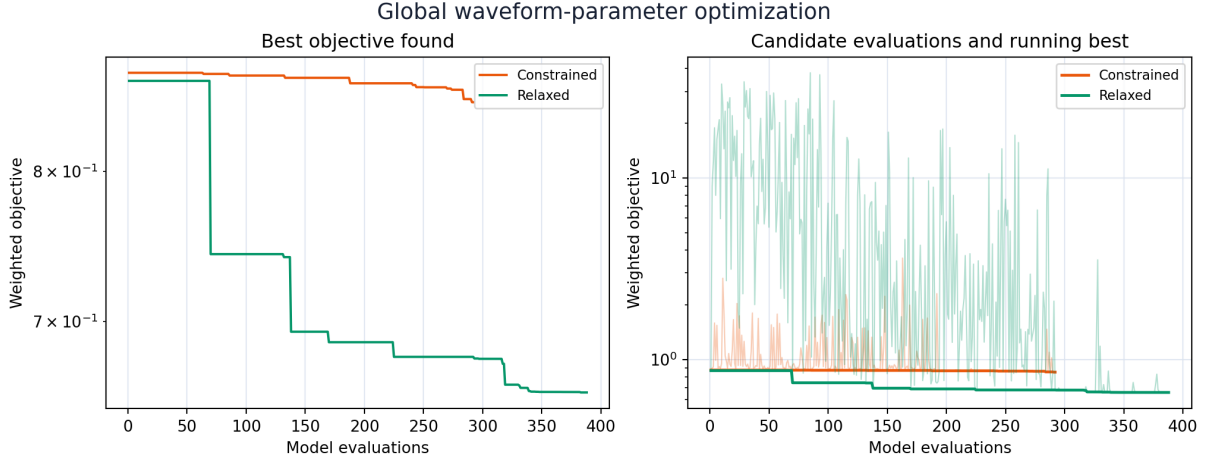


Figure 10: Optimization history for the physically constrained and relaxed searches. The large relaxed improvement occurs only after the search leaves the measured parameter region.

Model	Training J	Held-out J	All-trace J	Oscillations
Starting estimates	0.8680	0.9267	0.8813	0 of 22
Constrained fit	0.8601	0.9255	0.8750	0 of 22
Relaxed diagnostic	0.6515	0.7888	0.6827	0 of 22

Table 10: Fit and blind-test performance. Lower objective values are better. The experiment contains 11 oscillatory records.

The constrained fit lowers the all-trace objective by only 0.72% and the held-out objective by 0.13%. It remains essentially the independently estimated model and still predicts no oscillations. The relaxed search lowers the all-trace objective by 22.5% and the held-out objective by 14.9%, so its improvement is not confined to the training records. However, seven of its eight parameters lie outside their physical intervals, including $C = 5.42$ pF, and it also predicts no coherent oscillation. Its reduced amplitude error comes from large turn-on transients that settle to steady voltages.

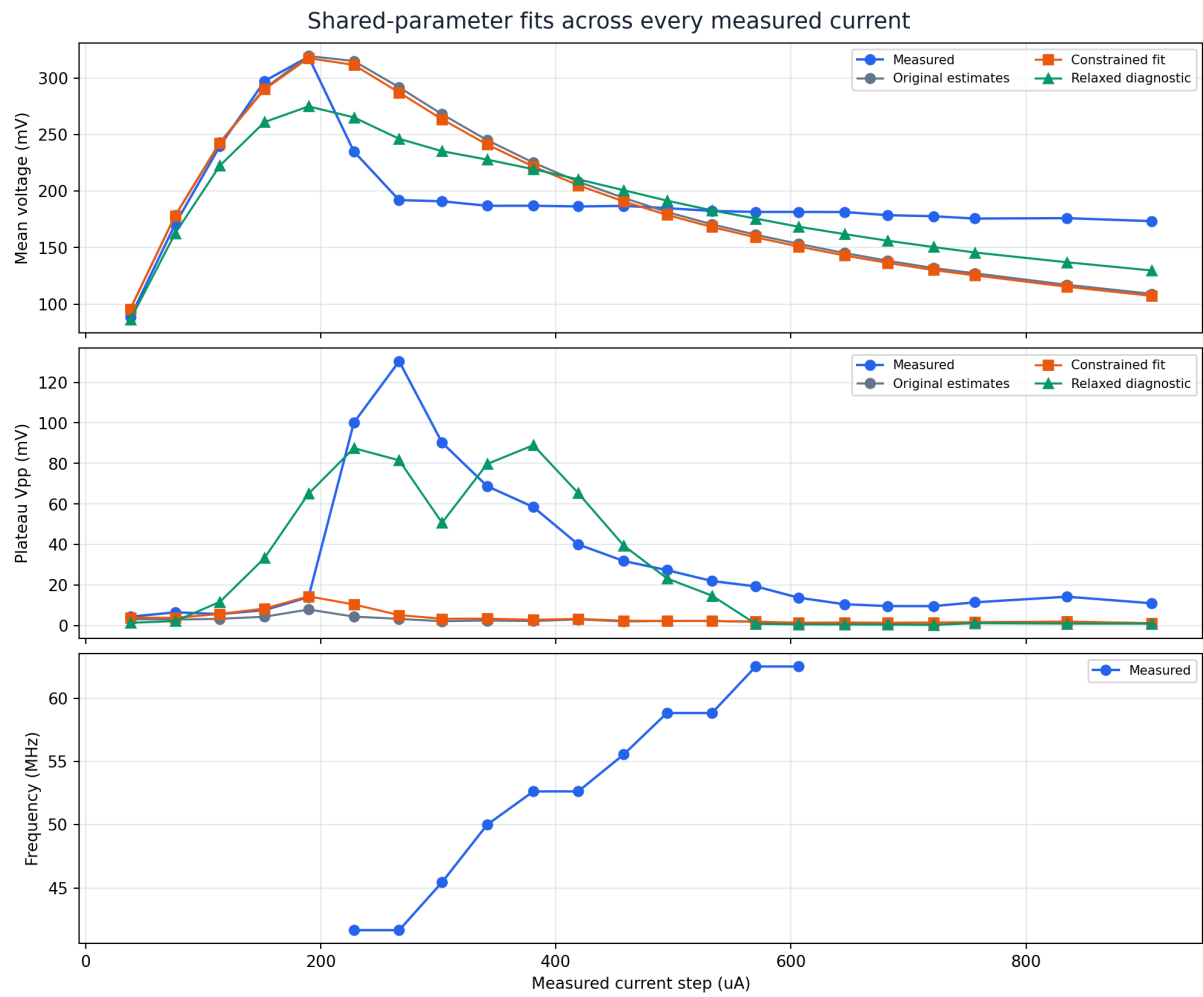


Figure 11: Measured and fitted operating summaries. The relaxed search partially reproduces mean voltage and apparent amplitude, but neither fit produces the measured frequency branch.

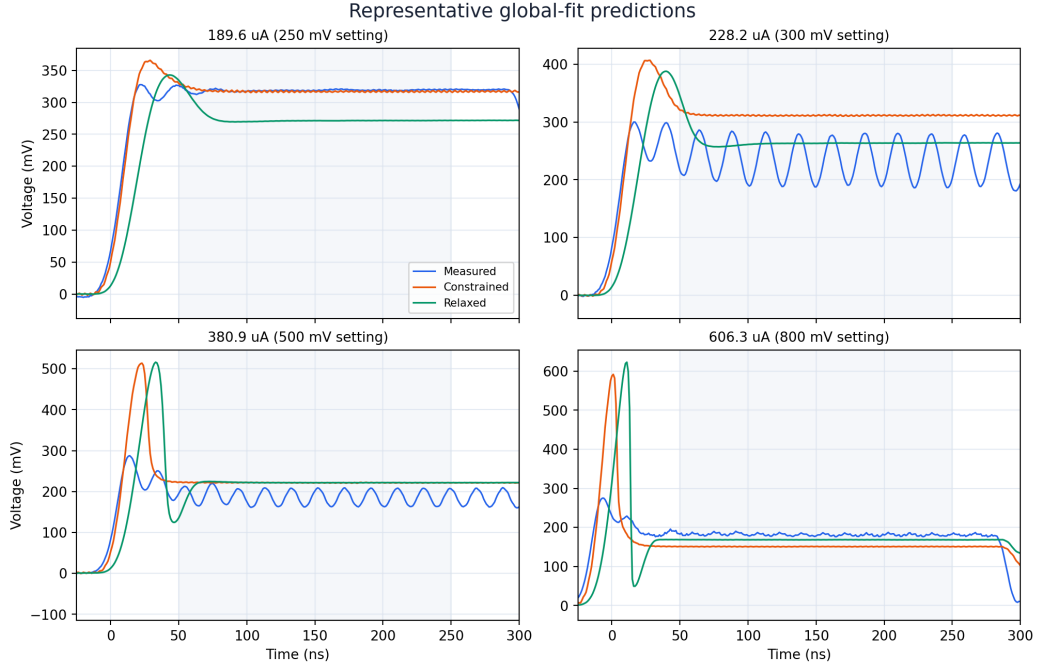


Figure 12: Representative waveform predictions. The relaxed fit can create a large initial excursion, but both fitted models settle instead of sustaining the measured oscillations.

Repeating the final evaluation at 0.2, 0.1, and 0.05 ns gives all-trace objectives of 0.6868, 0.6838, and 0.6827 for the relaxed fit, without changing any oscillation classification. The conclusion is therefore numerically converged. This exercise does not provide replacement physical parameter estimates: it shows that parameter adjustment within the present ideal-current, quasi-static-hysteresis model is insufficient. The likely missing information is dynamic switching behavior or the actual TIA/load impedance. Notably, γ remains near Yuanhang’s value in both searches, so changing the minor-loop curvature alone is not the solution.

10 Physics-anchored parameter inference

The unrestricted searches above showed that all eight parameters can compensate for one another. We therefore ordered the parameters by the strength of their independent evidence before fitting again. The measured $R(T)$ curve is the strongest anchor: R_0 , E_a/k_B , and R_m remain fixed at their fitted values, while T_c , w , and β may move only inside their bootstrap intervals. The measured ambient temperature is restricted to 314.25–314.55 K.

The environmental conductance is the next strongest dynamic estimate. At the final stable trace before oscillation, $dT/dt \simeq 0$, so

$$S_e = \frac{P}{T - T_0} = 0.003675 \text{ mW/K}, \quad 0.003434 \leq S_e \leq 0.004085 \text{ mW/K}. \quad (20)$$

This narrow interval is conditional on the 250-setting trace being quasi-steady and on the static heating branch applying to the driven device. The thermal capacitance is less precise, but the nonswitching heating traces still support $0.021918 \leq C_{\text{th}} \leq 0.092624$ pJ/K through $C_{\text{th}} = S_e \tau_{\text{th}}$. Only C and γ are searched broadly: the experiment provides an upper bound rather than a positive measurement for C , and no specimen-specific dynamic minor loops determine γ .

All measured oscillatory records are used during optimization. Two stable source settings, 50 and 1000 mV, are withheld as negative controls; this is not a blind test because the remaining records were already inspected. The fit is performed directly at $\Delta t = 0.025$ ns. Classification and false positives each receive weight 2500. Frequency receives weight 2, persistence and absolute amplitude weight 1, and the logarithmic amplitude-ratio penalty

$$J_{\text{relamp}} = \left[\ln \left(\frac{V_{\text{pp,pred}} + 3 \text{ mV}}{V_{\text{pp,meas}} + 3 \text{ mV}} \right) \right]^2 \quad (21)$$

receives weight 10. The remaining waveform, mean-voltage, spectrum, and edge terms act as secondary tie-breakers.

Parameter	Independent estimate	Allowed interval	Anchored fit
C (pF)	upper bound 0.390	0–30	6.83549*
C_{th} (pJ/K)	0.047873	0.021918–0.092624	0.044214
S_e (mW/K)	0.003675	0.003434–0.004085	0.003713
T_0 (K)	314.400	314.25–314.55	314.363
T_c (K)	333.494	333.137–333.838	333.338
w (K)	6.8823	6.6291–7.2681	7.0415
β (K ^{−1})	0.29916	0.28092–0.32021	0.31444
γ	0.95627 (Yuanhang)	0.1–3.0	0.15696*

Table 11: Physics-anchored shared fit. Asterisks mark the two weakly identified quantities that conflict with current support: C exceeds its timing bound and γ lies below the nominal 0.2–2.0 range.

Quantity	Experiment	Anchored fit
Peak-count classifications	22 reference labels	22 of 22
Peak-count oscillating records	11	11
Peak-count current interval	228.2–606.3 μA	228.2–606.3 μA
Stable negative controls	2	2
Median predicted/measured V_{pp}	1	$5.55\times$
V_{pp} mean absolute error	—	212.5 mV
Mean-voltage mean absolute error	—	35.4 mV
Frequency interval	41.7–62.5 MHz	32.3–125.0 MHz
Frequency mean absolute error	—	22.7 MHz

Table 12: Historical peak-count result at $\Delta t = 0.025$ ns. The later window audit finds that the 606.3 μA prediction decays; matching these labels does not establish the complete sustained oscillation window.

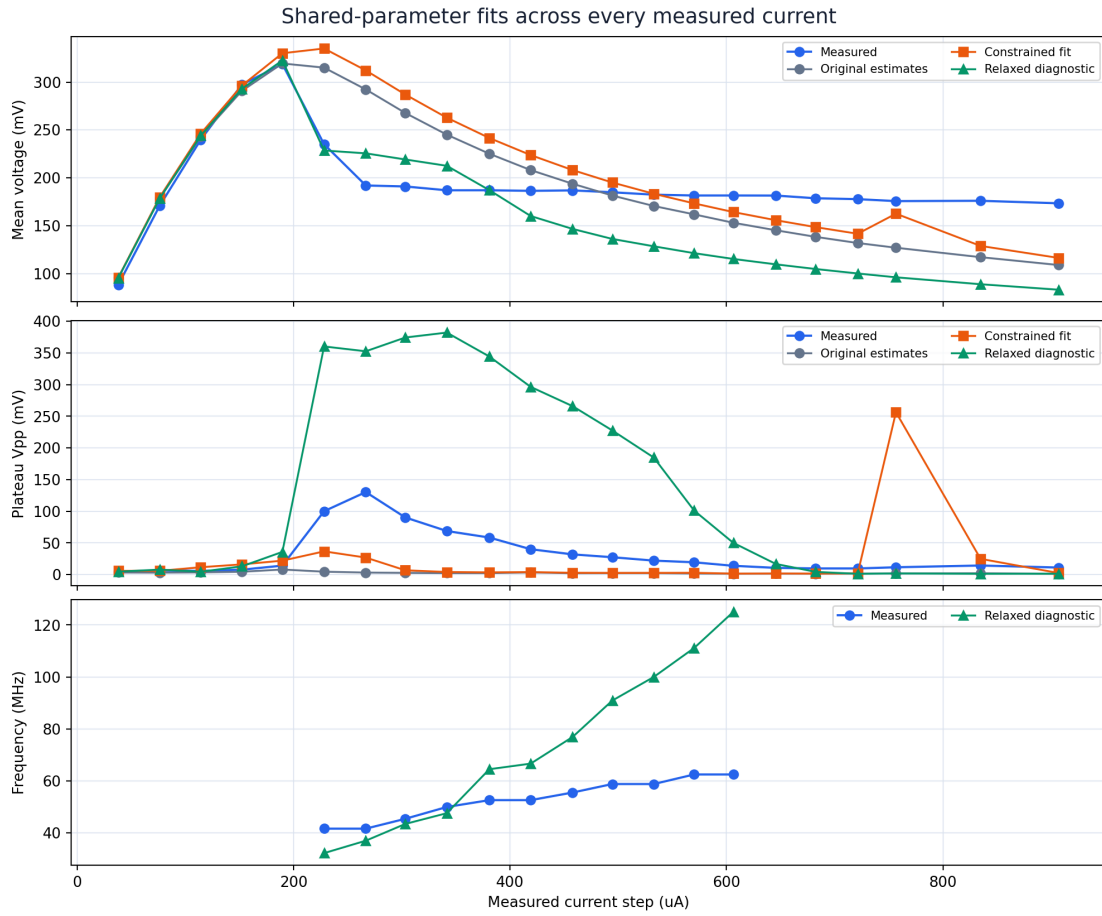


Figure 13: Measured and physics-anchored operating behavior under the historical detector. Amplitude is too large, frequency rises too rapidly, and the last detected oscillation is a decaying transient.

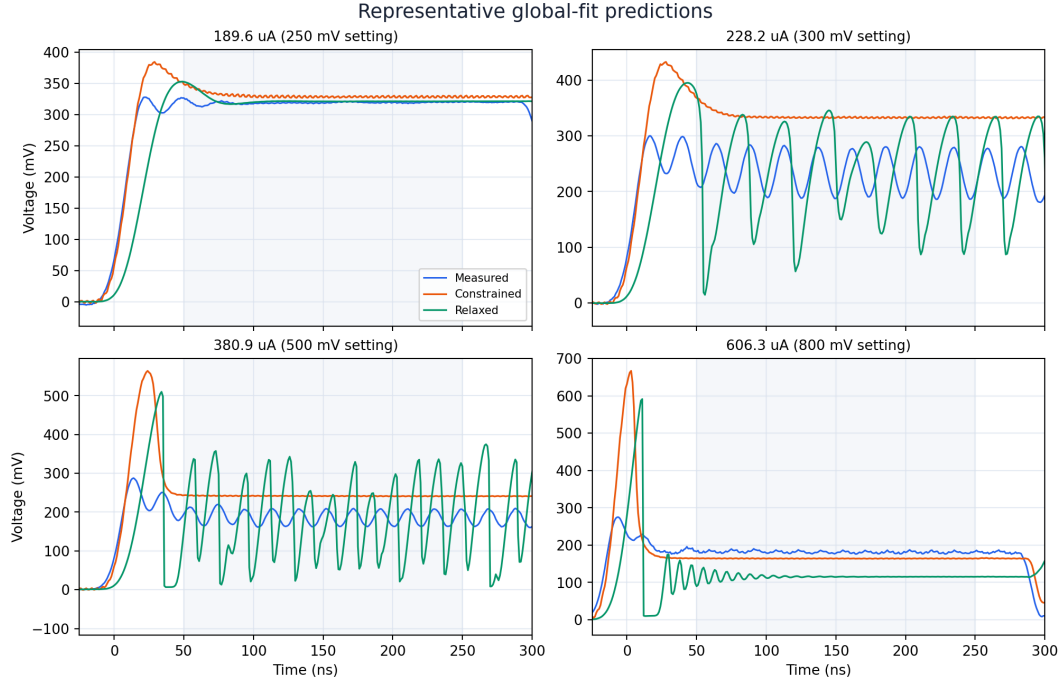


Figure 14: Representative physics-anchored predictions. The pre-onset record remains stable and several currents oscillate, but voltage amplitude is excessive and the 606.3 μA predicted cycles fade.

The same parameter vector gives 22/22 peak-count classifications at $\Delta t = 0.0125$ ns. It improved the earlier unrestricted classification-first result's 21/22 labels and reduced amplitude MAE from 280.1 to 212.5 mV while keeping six fitted quantities within their independent intervals. The earlier amplitude-priority vector reached 88.5 mV amplitude MAE, but only by falling to 19/22 classifications and moving seven quantities outside their intervals; it is not promoted as the specimen model.

The required $C = 6.84$ pF is about 17.5 times the experimental timing bound, and $\gamma = 0.157$ cannot be interpreted as a measurement without dynamic minor-loop data. The excessive amplitude follows the full fitted resistance excursion, approximately $I \Delta R$, while the frequency reaches 125 MHz at the upper boundary. These errors point to partial or spatially nonuniform switching and the actual TIA/load dynamics as hypotheses to test. The thermal estimates remain conditional on static $R(T)$ thermometry and circuit interpretation; satisfying their intervals does not establish that they are correct. The full-precision vector is archived because the classification basin is narrow; the printed digits are descriptive, not measurement precision. The optimizer reached its finite budget, so this is the best reproducible candidate found rather than proof of a unique global optimum.

11 Persistence audit and controlled parameter maps

The objective is to distinguish sustained experimental cycles from decaying ringing before searching for another parameter fit. We compare four consecutive 50 ns windows from 50 to 250 ns, measuring mean voltage, raw and 5–95% voltage spans, and fitted periodic amplitude. Late frequency is obtained independently from the 150–250 ns spectrum and refined by sinusoidal regression, without the old detector’s 8 ns minimum peak spacing. The 100 ns record still has approximately 10 MHz spectral bin spacing; numerical interpolation does not add measurement resolution.

A trace passes the operational persistence check when each window retains at least 6 mV fitted fundamental peak-to-peak amplitude, the last/first robust voltage-span ratio is at least 0.5, and the late fundamental explains at least 40% of detrended variance. These are diagnostic thresholds: varying the amplitude threshold from 4 to 8 mV and retention from 0.25 to 0.75 gives 9–13 measured detections, reflecting the small boundary signals. The nominal settings recover all 11 measured oscillators.

The archived 606.3 μA prediction falls from 50.2 mV raw peak-to-peak in 50–100 ns to 0.49 mV in 200–250 ns, while the experiment remains near 10–12 mV. Thus the earlier 22/22 peak-count result includes a dying oscillation.

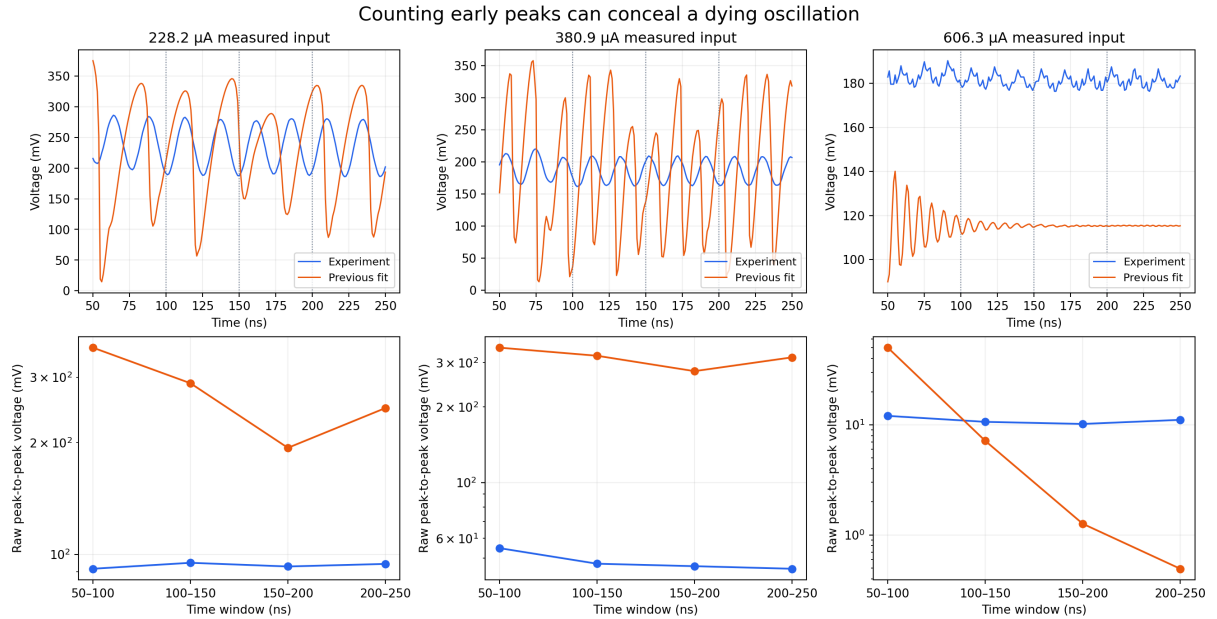


Figure 15: Window audit of the previous fit. The high-current experimental cycles persist while the model relaxes toward a steady voltage.

We then mapped 126 shared combinations of $C = 0\text{--}12$ pF and $\tau_{\text{th}} = C_{\text{th}}/S_e = 6\text{--}24$ ns at $\gamma = 0.157, 0.5$, and 0.956 , including the previous parameter vector. The other quantities remain fixed at that vector, including $S_e = 0.003713$ mW/K. The three measured inputs are approximately 228, 381, and 606 μA . All mapped C_{th} values remain within their conditional interval; $C > 0.39$ pF is explicitly a diagnostic stress test.

Candidates receive equal continuous penalties scaled to 20% amplitude ratio, 10% frequency error, 20 mV mean-voltage error, and a factor-two retention error. These are comparison scales, not statistical uncertainties. Fourteen combinations pass persistence at all three selected currents. We verify the best feature match, best persistent point, best timing-bound point, and previous

vector on all 22 currents at 0.025, 0.0125, and 0.00625 ns.

Candidate	C (pF)	τ_{th} (ns)	γ	Recovered	False+
Previous fit	6.835	11.909	0.157	9/11	0
Best feature match	8.000	13.026	0.157	10/11	0
Best persistent point	8.000	9.000	0.956	11/11	1
Best timing-bound point	0	11.909	0.157	0/11	0

Table 13: Persistence at the finest step, 0.00625 ns. Recovered counts refer to the 11 experimental oscillators; false positives are counted among the other records.

The persistent candidate additionally oscillates at 645.5 μA and its late 606.3 μA robust amplitude is 294.5 mV versus 9.25 mV measured ($31.8\times$). The best feature match still decays at 606.3 μA . The previous fit also fails the retention check at 570.1 μA at the two finest steps. No tested point with $C \leq 0.39$ pF sustains cycles at any of the three mapped currents.

Between 0.0125 and 0.00625 ns, the persistent candidate changes by at most 1.2% in late robust amplitude and 0.9% in frequency across its persistent currents. The best feature candidate changes by up to 17.3% and 13.7%, respectively. Matching labels alone therefore does not establish waveform convergence. No new specimen calibration is adopted: this conditional grid identifies a persistence–amplitude tradeoff without exhausting all parameter possibilities. The full four-feature maps are retained as supplementary numerical evidence.

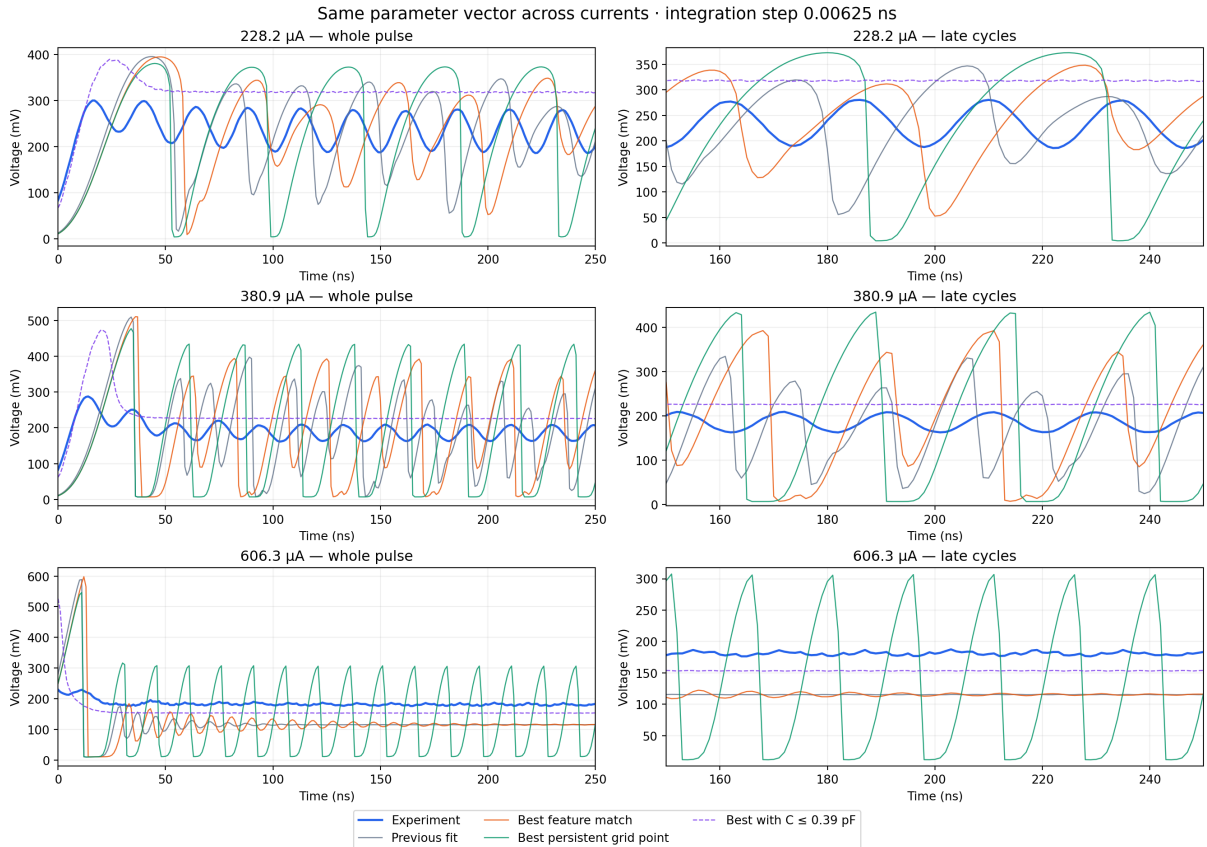


Figure 16: Shared candidates at 0.00625 ns. Preserving cycles can enlarge the voltage excursion, while the smaller-amplitude upper-boundary response still decays.

12 Conditional reconstruction of the driven resistance

To distinguish a search problem from inconsistent assumptions, we reconstruct the resistance and thermal history directly from the measured channels:

$$I_R = I - C \frac{dV}{dt}, \quad R_{\text{eff}} = \frac{V}{I_R}, \quad P = VI_R, \quad C_{\text{th}} \frac{dT}{dt} = P - S_e(T - T_0). \quad (22)$$

We then replay the existing hysteresis law on this prescribed $T(t)$ and compare its resistance with R_{eff} . Unlike a forward waveform fit, this check does not let an optimizer change the temperature trajectory to improve agreement. The reconstructed temperature is conditional on the thermal balance, parameters, and channel interpretation; it is not an independent temperature measurement. The static resistance curve is not used as a thermometer here.

The central calculation uses $C = 0.39$ pF, $S_e = 3.6751$ $\mu\text{W/K}$, $C_{\text{th}} = 0.047873$ pJ/K, $T_0 = 314.4$ K, the fitted major loop, and Yuanhang’s $\gamma = 0.95627$. Baseline-corrected channels are smoothed with a five-point quadratic filter before differentiating voltage. We integrate from -200 ns at T_0 , using piecewise-linear measured power, and compare the late 150–250 ns interval. Currents below are means in this interval after preprocessing, so they differ slightly from the earlier run labels.

Table 14: Late-time mean values from the conditional reconstruction.

I (μA)	Inferred T (K)	Measured R (Ω)	Replayed R (Ω)
189.6	330.91	1680.6	1679.2
228.7	328.96	1017.9	1762.5
381.1	333.64	488.4	1215.1
606.7	344.25	298.2	29.7

The below-onset agreement is a consistency check, not independent validation: this same operating region helped determine S_e . In contrast, the oscillatory records expose substantial resistance discrepancies, with opposite signs at intermediate and high current.

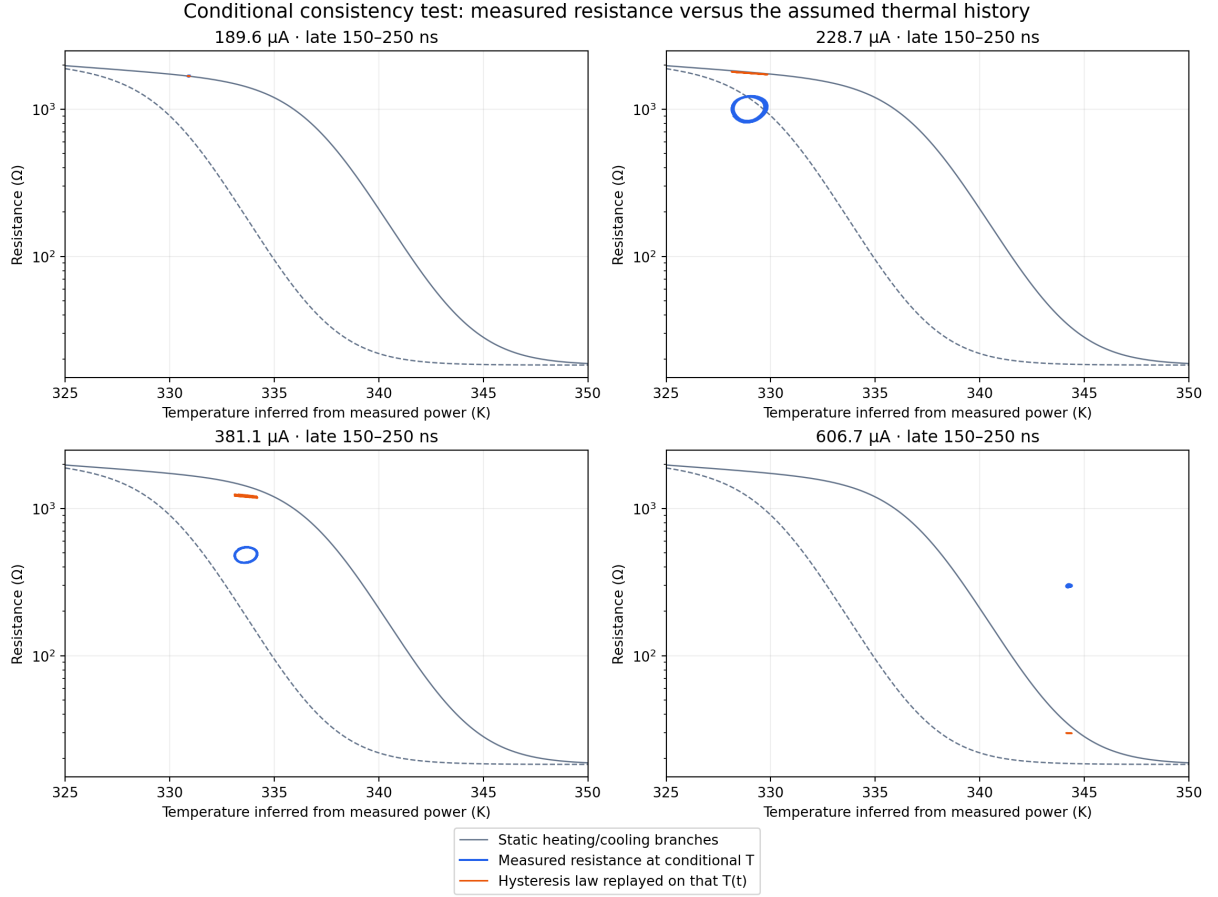


Figure 17: Measured effective resistance (blue) and the model resistance replayed on the inferred thermal history (orange), with the fitted static major loop in the background. Only the late interval is displayed; the replay includes the preceding history. These are conditional trajectories, not new forward fits.

All 22 records were tested at $\gamma = 0.15696, 0.5, 0.95627$, and 2. At $606.7 \mu\text{A}$, the central replay gives only $26.4\text{--}33.8 \Omega$ across these values, versus 298.2Ω measured. One-at-a-time checks use the previous thermal-parameter bounds, $C = 0$ and 6.8355 pF , and nine-point smoothing. The largest improvement at this current comes from the upper $S_e = 4.0850 \mu\text{W/K}$ bound: the inferred mean temperature falls to 341.26 K , but the replay still gives only $71.1\text{--}112.3 \Omega$ across the tested γ values. These checks are not a joint confidence region or an exhaustive parameter search. Refining the replay from 0.5 to 0.25 ns changes mean resistance by at most 4.85% across the tested cases and currents; it does not remove the large discrepancy. The underlying measured sampling remains 1 ns .

Thus, changing γ alone did not resolve this consistency test. The next useful step is to check the measured-channel interpretation and the transfer of the static resistance law and lumped thermal parameters to driven operation, before another broad fit. This evidence does not uniquely distinguish an incorrect thermal estimate, a circuit contribution, or missing phase dynamics. No new calibrated parameter set is adopted.

A Reproduction

The validation run is archived as

```
20260817_100102_current-step-with-a-nonzero-metallic-voltage-val_6765e0.
```

It can be reproduced from the repository root with

```
neuristor simulate current --config experiments/current/nonzero_voltage_valley.toml
```

The archived bundle contains the resolved configuration, traces, metrics, figure, and repository provenance. A fresh execution produced identical metrics and byte-identical trace data.

The resistance–temperature figure and synchronized GIF are regenerated from the same archived traces with `neuristor runs visualize`.

The sample resistance fit is archived as

```
20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9.
```

It is reproduced with

```
neuristor fit resistance --data data/experimental/100425_chip1_gap3.tsv --method
major-loop --bootstrap-samples 1000
```

The environmental-conductance analysis is archived as

```
20260817_153807_environmental-thermal-conductance-estimate_761640.
```

It is reproduced with

```
neuristor analyze conductance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--ambient-K 314.4 --ambient-interval-K 314.25,314.55
```

The thermal-capacitance analysis is archived as

```
20260828_112314_thermal-capacitance-estimate-with-conservative-0_aa2469.
```

It is reproduced with

```
neuristor analyze thermal-capacitance --data data/experimental/tia_current_sweep
--resistance-preset presets/resistance_100425_chip1_gap3.json
--resistance-bootstrap
public_jobs/20260816_125905_sample-r-t-major-loop-hysteresis-fit_0849a9/parameter_bootstrap.csv
--conductance-mW-per-K 0.003675126546984294
--conductance-bootstrap
public_jobs/20260817_153807_environmental-thermal-conductance-estimate_761640/conductance_bootstrap.csv
--ambient-K 314.4 --electrical-capacitance-pF 0.39
--selected-drives-mV 100,150,200 --fit-window-ns 15,35
```

The blind model comparison and capacitance study are archived as

```
20260829_100718_specimen-model-prediction-versus-measured-current_eefab7.
```

They are reproduced together with

```
neuristor analyze model-validation --config
experiments/current/specimen_model_validation.toml
```

The global shared-parameter inference is archived as

```
20260829_105704_global-specimen-parameter-inference-from-all-current_8f12d6.
```

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_waveform_inference.toml
```

The bundle contains the resolved objective and bounds, every fitted parameter, per-trace metrics and predictions, held-out validation, optimization histories, figures, and the three-step convergence check.

The physics-anchored inference is archived as

```
20260907_125315_physics-anchored-specimen-waveform-inference_717797.
```

It is reproduced with

```
neuristor analyze fit-waveforms --config
experiments/current/specimen_physics_anchored_inference.toml
```

Its bundle records the parameter-confidence hierarchy through explicit bounds, the amplitude-aware and oscillation-priority objective, both stable negative controls, all fitted waveforms and per-trace metrics, and the 0.1–0.0125 ns convergence audit. Earlier unrestricted oscillation and amplitude searches remain in the public archive as superseded development evidence.

The persistence audit and conditional reconstruction are archived as

```
20260914_073330_sustained-oscillation-audit-and-three-current-manifest_26ff13
20260914_074731_conditional-hysteresis-reconstruction-from-measured_fa4b66.
```

Their numerical workflows are reproduced with

```
neuristor analyze oscillation-audit --config
experiments/current/specimen_oscillation_audit.toml
neuristor analyze reconstruct-hysteresis --config
experiments/current/specimen_hysteresis_reconstruction.toml
```

The audit manifest separately records numerical and figure-rendering provenance. Both bundles include resolved inputs, numerical tables, figures, and methodology.

The historical capacitance shortcut is preserved for provenance in

20260817_134254_lab-current-trace-parameter-estimates_4223e0.

Its `electrical_capacitance.csv` contains the four values reconstructed in Table 6. The bundle is retained as evidence of how 19.8 pF was obtained, but that estimate is superseded by the full current-balance analysis in Section 5.

References

- [1] Y.-H. Zhang, C. Sipling, E. Qiu, I. K. Schuller, and M. Di Ventra, “Collective dynamics and long-range order in thermal neuristor networks,” *Nature Communications*, vol. 15, art. 6986, 2024. doi:10.1038/s41467-024-51254-4.
- [2] A. Gildor, S. Hodisan, S. Kvatinsky, and Y. Kalcheim, “Harnessing the VO₂ Phase Transition for Automatic Gain Control in Transimpedance Amplifiers,” manuscript, arXiv:2604.04594v1, 2026.